

Abirham Getie

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EDUCATION

The George Washington University

Master of Science, Data Science

Washington, DC

May 2026

Relevant Coursework: Machine Learning, Data Mining, Big Data Analytics, Data Warehousing, Cloud Computing (AWS), NLP

Gujarat Technological University

Bachelor of Engineering, Computer Engineering

Ahmedabad, India

June 2024

EXPERIENCE

Ethio Telecom

Data Analyst

Addis Ababa, Ethiopia

Jan 2022 – Jun 2024

- Optimized database indexing on 500K+ customer service logs and network records, cutting query response time by 20% and accelerating weekly IT operations reporting workflows.
- Built a Python anomaly detection system that flagged IT failures from streaming network logs, replacing manual log review and shifting the team from reactive to proactive incident response.
- Built interactive Tableau dashboards visualizing network outage hotspots, giving the IT team a single source of truth for triaging field issues.

Open Avenues Foundation, Build Fellowship

AI/ML Fellow (Part-time)

Remote (New York, NY)

Feb 2025 – May 2025

- Built an AI-powered image retrieval system on Caltech-101 under industry mentorship, extracting 2048-dim embeddings from a pretrained ResNet-50 and indexing with FAISS for fast nearest-neighbor search.
- Engineered an OpenCV image quality pipeline (blur and JPEG-artifact detection) to filter corrupted images before indexing; validated with t-SNE and deployed a Docker-containerized Streamlit app for real-time visual search.

PROJECTS

AI-Powered IT Ticket Triage & Knowledge Base Generation | GW IT Capstone [GitHub](#)

Python, FastAPI, React, TypeScript, PostgreSQL, pgvector, Azure OpenAI (GPT-4.1, ada-002), AWS EC2, Docker

- Built a production-grade full-stack RAG application on 100K+ enterprise records (68K+ IT support tickets and 40K+ work orders), with a FastAPI backend, React/Next.js frontend, and AWS EC2 deployment, automating knowledge work for an enterprise IT team.
- Designed an agentic LLM system orchestrating GPT-4.1 across labeling, summarization, deduplication, and LLM-as-judge validation; generated 40 validated KB articles from 73 critical gaps with 3-stage quality gates.
- Built a semantic search and clustering pipeline using ada-002 embeddings (1,536-dim) and pgvector, with custom silhouette scoring identifying 175 distinct topic clusters.

Financial News Sentiment Classification [GitHub](#)

Python, PyTorch, Hugging Face Transformers, Streamlit

- Built an end-to-end NLP pipeline classifying 1.8M financial headlines, validating FinBERT pseudo-labels against the expert-annotated Financial PhraseBank with 97.2% agreement.
- Benchmarked 5 models (LogReg, BiLSTM, DistilBERT, BERT-Base, FinBERT) plus a soft-voting ensemble; FinBERT achieved 95.7% test accuracy and 0.96 weighted F1, beating the classical baseline by 15 points.

TECHNICAL SKILLS

LLMs & GenAI: GPT-4, Claude, RAG, Agentic Systems, LLM-as-Judge, Prompt Engineering, Fine-tuning, Hugging Face, OpenAI API, Embeddings

ML, CV & NLP: PyTorch, scikit-learn, BERT, DistilBERT, FinBERT, ResNet-50, FAISS, OpenCV, t-SNE, K-Means, NLTK, spaCy

Full Stack & Cloud: Python, TypeScript, SQL, FastAPI, React, REST APIs, PostgreSQL, MongoDB, AWS, Docker, GitHub Actions

PUBLICATIONS

Amharic Hate Speech Detection, Springer book chapter (2024) · *Fake News Identification*, Journal of Engineering Science (2024).